A.3 Chapter 11: Scaling Gaussian processes to large data sets

This exercise demonstrates the improvement in efficiency when going from a regular GP model to a VGP one on a real-life data set of housing prices in California. Our goal is to once again observe the computational benefits of a VGP, this time in a real-world setting.

There are multiple steps:

1. We use the Pandas library to read in the data set:

```
import pandas as pd

df = pd.read_csv("../data/housing.csv")
```

   Once read in, the Pandas dataframe should look similar to the output in Figure A.3.

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
0	-117.08	32.70	37	2176	418.0	1301	375	2.8750	98900
1	-117.91	34.11	20	3158	684.0	2396	713	3.5250	153000
2	-117.10	32.75	11	2393	726.0	1905	711	1.3448	91300
3	-117.22	32.74	52	1260	202.0	555	209	7.2758	345200
4	-121.99	37.29	32	2930	481.0	1336	481	6.4631	344100
...	...	...	...	...	...	...	...	...	...
4995	-118.41	34.19	42	779	145.0	450	148	3.9792	193800
4996	-122.96	38.42	50	2530	524.0	940	361	2.9375	122900
4997	-118.00	33.77	24	1324	267.0	687	264	3.4327	192800
4998	-122.81	38.54	12	2289	611.0	919	540	1.1553	139300
4999	-118.08	33.84	25	3696	953.0	2827	860	3.3438	153300

Figure A.3 The housing price data set shown as a Pandas dataframe. This is the training set for this exercise.

2. We create the scatter plot as follows:

```
plt.figure(figsize=(8, 6))
plt.scatter(df.longitude, df.latitude, c=np.log(df.median_house_value))
plt.colorbar();
```

   The visualization should look similar to Figure A.4.

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